EASY · ALGEBRA · ANSWER KEY

Linear equations in two variables

01

10 answers · Rationale-rich · Teach from doc

ANSWER KEY · SAT QUESTION BANK

2026-05-29

Q1

D

D. $x + 3y = 70$

Choice D is correct. Since x represents the number of 1-point questions and y represents the number of 3-point questions, the assignment is worth a total of $1 \cdot x + 3 \cdot y$, or $x + 3y$, points. Since the assignment is worth 70 points, the equation $x + 3y = 70$ represents this situation.

Choice A is incorrect and may result from conceptual errors.

Choice B is incorrect and may result from conceptual errors.

Choice C is incorrect and may result from conceptual errors.

Q2

D

D. $y = \frac{1}{9}x + 14$

Choice D is correct. The equation of a line in the xy-plane can be written as $y = mx + b$, where m represents the slope of the line and $0, b$ represents the y-intercept of the line. It's given that the slope of the line is $\frac{1}{9}$. It follows that $m = \frac{1}{9}$. It's also given that the line passes through the point $0, 14$. It follows that $b = 14$. Substituting $\frac{1}{9}$ for m and 14 for b in $y = mx + b$ yields $y = \frac{1}{9}x + 14$. Thus, the equation $y = \frac{1}{9}x + 14$ represents this line.

Choice A is incorrect. This equation represents a line with a slope of $-\frac{1}{9}$ and a y-intercept of $0, -14$.

Choice B is incorrect. This equation represents a line with a slope of $-\frac{1}{9}$ and a y-intercept of $0, 14$.

Choice C is incorrect. This equation represents a line with a slope of $\frac{1}{9}$ and a y-intercept of $0, -14$.

Q3

14

The correct answer is 14. It's given that the equation $46 = 2a + 2b$ gives the relationship between the side lengths a and b of a certain parallelogram. Substituting 9 for a in the given equation yields $46 = 29 + 2b$, or $46 = 18 + 2b$. Subtracting 18 from both sides of this equation yields $28 = 2b$. Dividing both sides of this equation by 2 yields $14 = b$. Therefore, if $a = 9$, the value of b is 14.

Q4

A

A. $3w + 5r = 14$

Choice A is correct. Since Jay walks at a speed of 3 miles per hour for w hours, Jay walks a total of $3w$ miles. Since Jay runs at a speed of 5 miles per hour for r hours, Jay runs a total of $5r$ miles. Therefore, the total number of miles Jay travels can be represented by $3w + 5r$. Since the combined total number of miles is 14, the equation $3w + 5r = 14$ represents this situation.

Choice B is incorrect and may result from conceptual errors.

Choice C is incorrect and may result from conceptual errors.

Choice D is incorrect and may result from conceptual errors.

Q5

A

A.

x	y
0	8
2	148
4	288

Choice A is correct. Each of the given choices gives three values of x: 0, 2, and 4. Substituting 0 for x in the given equation yields $y = 70(0) + 8$, or $y = 8$. Therefore, when $x = 0$, the corresponding value of y for the given equation is 8. Substituting 2 for x in the given equation yields $y = 70(2) + 8$, or $y = 148$. Therefore, when $x = 2$, the corresponding value of y for the given equation is 148. Substituting 4 for x in the given equation yields $y = 70(4) + 8$, or $y = 288$. Therefore, when $x = 4$, the corresponding value of y for the given equation is 288. Thus, if the three values of x are 0, 2, and 4, then their corresponding values of y are 8, 148, and 288, respectively, for the given equation.

Choice B is incorrect. This table gives three values of x and their corresponding values of y for the equation $y = 4x + 70$.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q6

B

B. $y = -\frac{1}{2}x + 3$

Choice B is correct. A line in the xy-plane with a slope of m and a y-intercept of $0, b$ can be represented by the equation $y = mx + b$. It's given that the line has a slope of $-\frac{1}{2}$. Therefore, $m = -\frac{1}{2}$. It's also given that the line passes through the point $0, 3$. Therefore, $b = 3$. Substituting $-\frac{1}{2}$ for m and 3 for b in the equation $y = mx + b$ yields $y = -\frac{1}{2}x + 3$. Therefore, the equation $y = -\frac{1}{2}x + 3$ represents this line.

Choice A is incorrect. This equation represents a line in the xy-plane that passes through the point $0, -3$, not $0, 3$.

Choice C is incorrect. This equation represents a line in the xy-plane that has a slope of $\frac{1}{2}$, not $-\frac{1}{2}$, and passes through the point $0, -3$, not $0, 3$.

Choice D is incorrect. This equation represents a line in the xy-plane that has a slope of $\frac{1}{2}$, not $-\frac{1}{2}$.

Q7

B

B. $y = 7x + 5$

Choice B is correct. The equation of a line in the xy-plane can be written in slope-intercept form $y = mx + b$, where m is the slope of the line and $0, b$ is its y-intercept. It's given that the line passes through the point $0, 5$. Therefore, $b = 5$. It's also given that the line is parallel to the graph of $y = 7x + 4$, which means the line has the same slope as the graph of $y = 7x + 4$. The slope of the graph of $y = 7x + 4$ is 7. Therefore, $m = 7$. Substituting 7 for m and 5 for b in the equation $y = mx + b$ yields $y = 7x + 5$.

Choice A is incorrect. The graph of this equation passes through the point $0, 0$, not $0, 5$, and has a slope of 5, not 7.

Choice C is incorrect. The graph of this equation passes through the point $0, 0$, not $0, 5$.

Choice D is incorrect. The graph of this equation passes through the point $0, 7$, not $0, 5$, and has a slope of 5, not 7.

Q8

A

A. $y = 6x + 8$

Choice A is correct. The slope-intercept form of an equation for a line is $y = mx + b$, where m is the slope of the line and b is the y-coordinate of the y-intercept of the line. It's given that the slope is 6, so $m = 6$. It's also given that the line passes through the point $(0,8)$ on the y-axis, so $b = 8$. Substituting $m = 6$ and $b = 8$ into the equation $y = mx + b$ gives $y = 6x + 8$.

Choices B, C, and D are incorrect and may result from misinterpreting the slope-intercept form of an equation of a line.

Q9

.25

Accept also: 1/4

The correct answer is $\frac{1}{4}$. It's given that line k is defined by $y = \frac{1}{4}x + 1$. It's also given that line j is parallel to line k in the xy-plane. A line in the xy-plane represented by an equation in slope-intercept form $y = mx + b$ has a slope of m and a y-intercept of $0, b$. Therefore, the slope of line k is $\frac{1}{4}$. Since parallel lines have equal slopes, the slope of line j is $\frac{1}{4}$. Note that 1/4 and .25 are examples of ways to enter a correct answer.

Q10 A

A. 5

Choice A is correct. Let x represent the number of rabbit snails that Naomi bought. It's given that each rabbit snail costs $\$8$. Therefore, the total cost, in dollars, of the rabbit snails that Naomi bought can be represented by the expression $8x$. It's also given that each nerite snail costs $\$6$, and that Naomi bought 2 nerite snails. Therefore, the total cost, in dollars, of the nerite snails that Naomi bought is 62, or 12. Since Naomi bought both the rabbit snails and the nerite snails for a total of $\$52$, the equation $8x + 12 = 52$ can be used to represent the situation. Subtracting 12 from both sides of this equation yields $8x = 40$. Dividing both sides of this equation by 8 yields $x = 5$. Therefore, Naomi bought 5 rabbit snails.

Choice B is incorrect. This is the total cost, in dollars, of the nerite snails that Naomi bought, not the number of rabbit snails.

Choice C is incorrect. This is the cost, in dollars, of one rabbit snail and one nerite snail, not the number of rabbit snails that Naomi bought.

Choice D is incorrect and may result from conceptual or calculation errors.

Notes

Accept equivalent forms (e.g. `0.25 ≡ 1/4`). For grid-in items, decimal and fraction answers are both valid.

Generated 2026-05-29 · cb-educator-question-bank